

Qingyu Xiao

Robotics PhD Candidate | Georgia Institute of Technology

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[Website](#) | [Google Scholar](#) | [LinkedIn](#)

Profile

Robotics PhD candidate advised by [Prof. Matthew Gombolay](#). Works published in ICRA and IROS.

- **Sense:** Computer vision, multi-sensor fusion, real-time multi-camera localization, and camera-LiDAR cuboid detection; **C++17, ROS 2, and OpenCV**.
- **Think:** Probabilistic state estimation, factor-graph optimization, and differentiable dynamics learning; **GTSAM, Python, and PyTorch**.
- **Act:** Learning from demonstrations and diffusion-based motion planning for robot striking and wheelchair navigation.

Current ongoing research: **STAG (Sense, Think, Action Graph)** with [Dr. Frank Dellaert](#), jointly modeling perception, prediction, and action under uncertainty.

Education

Georgia Institute of Technology

PhD Candidate, Robotics

2021 – Dec. 2027 (expected)

Atlanta, GA

Duke University

MS, Mechanical Engineering and Materials Science

2016 – 2018

Durham, NC

North China Electric Power University

BS, Energy and Power Engineering; Outstanding Engineers Class

2012 – 2016

Baoding, China

Research Experience

01 / Learn to Act Agile robot behavior

Egocentric Perception to Robot Action

Ongoing

- Explore **factor-graph modeling** to connect egocentric visual observations, state prediction, and robot actions for tennis interaction on the [ESTHER](#) platform.
- Investigate **joint probabilistic learning** through STAG with Dr. Frank Dellaert, modeling dependencies and uncertainty across sensing, prediction, and action.

Diverse Robot Striking Motions

ICRA 2025

- Contributed to **learning from demonstrations** with **diffusion models** for robot striking; incorporated kinematic constraint guidance to improve generated trajectory accuracy. [Paper](#)

Wheelchair Tennis Navigation

ICRA 2025

- Contributed to **diffusion-based motion planning** learned from broadcast videos; applied domain knowledge transfer to generate smooth wheelchair navigation trajectories. [Paper](#)

02 / Learn to Sense Robot perception and localization

Real-Time Robot Localization from Multi-Camera Line Landmarks

IROS 2026; accepted

- Built a **multi-camera sensor-fusion** pipeline for asynchronous robot pose estimation using visual line landmarks, factor-graph optimization, and **incremental GTSAM inference**.
- Engineered a **C++17/ROS 2** pipeline with composable nodes and zero-copy intra-process communication, achieving **160 FPS** and approximately **10 cm localization error** without LiDAR.
- Applied **ControlNet-based synthetic data generation** to augment training data for visual line segmentation.

Research Experience (continued)

03 / Learn to Think Dynamics learning and state prediction

Learning Ball Dynamics with Differentiable Factor Graphs

ICRA 2025

- Built an end-to-end framework combining **differentiable optimization** with **neural dynamics learning** to jointly train a factor-graph state estimator and predictor from camera observations.
- Applied **Gram-Schmidt representations** to encode roto-translational invariance in flight and bounce models, improving sample efficiency relative to data augmentation.
- Designed **self-multiplicative network connections** to improve nonlinear dynamics prediction. [Paper](#)

Asynchronous Multi-Camera Ball Localization and Prediction

ICRA 2024

- Applied **multi-camera tracking** and **human pose estimation** to fuse asynchronous ball observations and pose-derived spin priors in a probabilistic factor graph.
- Implemented **incremental state estimation in GTSAM** at approximately **5 ms** per inference; reduced landing-position prediction error by **63.6%** against an adaptive extended Kalman filter baseline. [Paper](#)

Industry Experience

Research Intern, Stack AV

Summer 2025

Pittsburgh, PA

- Developed **camera-LiDAR fusion** methods for **3D cuboid detection** in autonomous driving using learnable factor graphs.
- Applied **probabilistic graphical modeling** to derive sensor-conditioned cuboid distributions and combine evidence from multiple sensing modalities.
- Improved recall from approximately **50% to 90%** while maintaining precision around **90%**.

Computer Vision Intern, SkyMul

Summer 2022

Peachtree Corners, GA

- Developed **RGB-D perception and 6D pose estimation** for rebar intersections, integrating real-time detection with a Unitree A1 robot for on-site rebar tying.
- Implemented **GPU-accelerated Gaussian mixture modeling** to estimate intersection poses from 3D point-cloud distributions.
- Integrated rebar-intersection landmarks into **visual-inertial SLAM** using **GTSAM factor graphs**.

Technical Skills

Programming: Python, C++17

Estimation: GTSAM, factor-graph optimization, multi-sensor fusion, visual-inertial SLAM

Vision: OpenCV, YOLO, 3D pose estimation, point-cloud processing, camera-LiDAR fusion

Learning: PyTorch, differentiable optimization, neural dynamics models, diffusion models

Robotics: ROS/ROS 2, Isaac Sim, Isaac Lab, real-time robot integration

Systems: Linux, Git, Docker, CMake, Anaconda, uv

Platforms and sensors: Franka Emika and WAM robot arms, IMUs, RGB-D cameras, high-speed cameras, multi-camera systems

Teaching Experience

Teaching Assistant, CS 6601: Artificial Intelligence

2023 – 2024

Georgia Institute of Technology

Selected Publications

Full publication list on [Google Scholar](#)

1. **Real-Time Robot Localization from Multi-Camera Line Landmarks.** IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026. *Accepted*.
2. **Qingyu Xiao**, Zixuan Wu, and Matthew Gombolay. [Learning Dynamics of a Ball with Differentiable Factor Graph and Roto-Translational Invariant Representations.](#) IEEE International Conference on Robotics and Automation (ICRA), 2025.
3. Kin Man Lee, Sean Ye, **Qingyu Xiao**, et al. [Learning Diverse Robot Striking Motions with Diffusion Models and Kinematically Constrained Gradient Guidance.](#) IEEE International Conference on Robotics and Automation (ICRA), 2025.
4. Zixuan Wu, Zulfiqar Zaidi, Adithya Patil, **Qingyu Xiao**, and Matthew Gombolay. [Learning Wheelchair Tennis Navigation from Broadcast Videos with Domain Knowledge Transfer and Diffusion Motion Planning.](#) IEEE International Conference on Robotics and Automation (ICRA), 2025.
5. **Qingyu Xiao**, Zulfiqar Zaidi, and Matthew Gombolay. [Multi-Camera Asynchronous Ball Localization and Trajectory Prediction with Factor Graphs and Human Poses.](#) IEEE International Conference on Robotics and Automation (ICRA), 2024.
6. Esmaeil Seraj, Kin Man Lee, Zulfiqar Zaidi, **Qingyu Xiao**, et al. [Interactive and Explainable Robot Learning: A Comprehensive Review.](#) *Foundations and Trends in Robotics*, 2024.

Academic Service

Workshop Organizer, RoboLetics 3.0 Benchmarks for Agile Physical Human-Robot Interaction.	CoRL 2026
Session Chair and Co-Chair Chair: Vision Systems for Robotics and Industrial Applications. Co-Chair: Visual Place Recognition Through Novel Sensing.	IROS 2026
Workshop Organizer, RoboLetics 2.0 Athletic Robots and Dynamic Motor Skills.	ICRA 2025

Honors and Awards

- Honorable Mention, ASME Journal of Mechanisms and Robotics Best Paper Award, 2023.
- First Place, ASME-CIE Hackathon, August 2022.
- Third Place, ASME-CIE Hackathon, August 2021.